SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING) Department of Computer Science and Engineering			
CO4 ASSESSMENT TOOL 1 – Test Case Design Assignment			
Course Code:	CSA10 / CSA1008	**Course Title:**	Software Engineering
CO Assessed:	CO4	**Assessment:**	Assessment Tool 1 – Test Case Design Assignment
Weightage:	20%	**Total Marks:**	25
CO4 Statement:	Implement robust testing, quality assurance, and DevOps practices, emphasizing security, reliability, and ethics.		
Student Name:	Chodam Nisha	**Reg. No.:**	192324306
Date:	19-08-2026	**Duration:**	60 minutes

Code of Conduct: *I* **Chodam Nisha (192324306)** *certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: **Chodam Nisha**

1. PROBLEM STATEMENT & SYSTEM DESCRIPTION

The **IoT-Based Flood Early Warning System (FEWS)** is a critical public safety infrastructure platform designed to track, analyze, and manage real-time hydrological hazards in river basins and coastal zones. The system continuously ingests telemetry data—including water levels (meters), precipitation intensity (mm/hr), and dam discharge flow rates (m^3/s)—from ultrasonic sensors, automated rain gauges, and flow meters every 10 seconds via encrypted MQTT/HTTPS protocols.

FEWS addresses delayed evacuation responses, uncoordinated barrier deployments, and inaccurate threat assessments by monitoring live sensor streams against multi-tiered safety thresholds (Normal, Advisory Alert, Warning, and Evacuation), executing automated siren actuation and flood barrier gate controls to mitigate damage, and calculating dynamic flood arrival times and spatial threat zones to facilitate targeted emergency communications.

2. REQUIREMENT IDENTIFICATION (FUNCTIONAL & NON-FUNCTIONAL)

Req ID	Type	Requirement Name	Requirement Description & Business Rule
FR-01	Functional	User Authentication & RBAC	Authenticate Emergency Officials and Public Users with MFA, enforcing role-based dashboard and control access.
FR-02	Functional	IoT Telemetry Ingestion	Ingest water level (m), rain rate (mm/hr), and flow rate (m^3/s) from IoT sensors every 10 seconds via MQTT/HTTPS.
FR-03	Functional	Threshold & Flood Alerting	Trigger alerts: Advisory Alert (80% bank capacity), Warning Level (90%), and Critical Evacuation Level ($\geq 100\%$).
FR-04	Functional	Flash Flood Detection	Detect rapid water rise (>0.5 m/min) or heavy rainfall (>50 mm/hr) to immediately flag critical flash flood status.
FR-05	Functional	Automated Barrier Gate Control	Send automated or manual operational commands to IoT flood barrier gates within 3 seconds of trigger/user confirmation.
FR-06	Functional	Zone Threat Mapping	Calculate spatial risk scores and dispatch targeted SMS alerts based on threat zones (Zone 1: High, Zone 2: Moderate, Zone 3: Low).
NFR-01	Non-Functional	Telemetry Latency	IoT sensor data ingestion to dashboard visualization and alert dispatch latency must not exceed 2.0 seconds.

Req ID	Type	Requirement Name	Requirement Description & Business Rule
NFR-02	Non-Functional	Security & Encryption	All sensor packets, API endpoints, and user credentials must be encrypted via TLS 1.3 and AES-256 standards.
NFR-03	Non-Functional	System Uptime	The cloud monitoring platform must maintain 99.99% uptime with offline telemetry buffer synchronization during power outages.

3. TEST SCENARIOS & APPLIED TEST DESIGN TECHNIQUES

Scen ID	Req ID	Test Scenario Description	Technique Applied	Technique Justification & Partition
TS-01	FR-01	Verify Admin Authentication & MFA Validation	Equivalence Partitioning & BVA	Valid admin credentials vs invalid length/format and expired MFA tokens.
TS-02	FR-02	Verify IoT Water Level Telemetry Ingestion	Boundary Value Analysis (BVA)	Valid sensor range (0.0m to 20.0m). Boundaries: -0.1m, 0.0m, 0.1m, 19.9m, 20.0m, 20.1m.
TS-03	FR-03	Verify Flood Risk Alert Threshold Triggers	Equivalence Partitioning & BVA	Partitions: Normal (<80%), Advisory (80%-89%), Warning (90%-99%), Evacuation ($\geq 100\%$).
TS-04	FR-04	Verify Flash Flood Detection Algorithm	Decision Table Testing	Rule-based matrix combining rise rate, rainfall intensity, duration, and dam status.
TS-05	FR-05	Verify Barrier Gate Actuation & State Changes	State Transition Testing	States: CLOSED → OPENING → OPEN → ERROR → MANUAL_OVERRIDE → CLOSED.
TS-06	FR-06	Verify Threat Zone Alert Dispatch Matrix	Equivalence Partitioning & BVA	Zone 1 (0-5km radius), Zone 2 (5-15km radius), Zone 3 (>15km radius).
TS-07	NFR-02	Verify API Endpoint Security & SQLi Prevention	Error Guessing & Security Testing	Malicious SQL/XSS payloads in API parameters and header injection.

4. DETAILED TEST CASES (NORMAL, BOUNDARY, INVALID, EXCEPTIONAL)

TC ID	Scen ID	Test Condition / Steps	Test Data	Expected Result	Actual Result	Status
TC-001	TS-01	Normal: Enter valid registered admin credentials & MFA code.	User: 'admin@fews.gov' Pass: 'Safety#2026' MFA: '849201'	Login successful; JWT admin token generated; redirected to Control Room UI.	Logged in successfully and redirected.	**Pass**
TC-002	TS-01	Boundary/Invalid: Password length below minimum (7 chars).	User: 'admin@fews.gov' Pass: 'Fwd#123'	Validation error: 'Password must be 8-20 characters long.' Form blocked.	Validation error displayed as expected.	**Pass**
TC-003	TS-02	Normal: Ingest valid water level telemetry packet via MQTT.	Device: 'WLS-301' Level: 4.5 m Time: '2026-08-19 10:00'	Telemetry parsed; DB updated; dashboard updates in <2.0s.	DB updated and UI refreshed in 1.1s.	**Pass**
TC-004	TS-02	Boundary: Ingest lower boundary valid level (0.0 m - dry bed).	Device: 'WLS-301' Level: 0.0 m	Recorded as normal dry bed state (0.0m); baseline maintained.	Recorded 0.0m successfully.	**Pass**
TC-005	TS-02	Boundary/Invalid: Ingest negative water level (-0.5 m).	Device: 'WLS-301' Level: -0.5 m	Packet rejected; error logged: 'Invalid negative level reading'.	Packet rejected with HTTP 422.	**Pass**

TC ID	Scen ID	Test Condition / Steps	Test Data	Expected Result	Actual Result	Status
TC-006	TS-02	Exceptional: Ingest level exceeding physical max limit (22.0 m).	Device: 'WLS-301' Level: 22.0 m	Logged as 'Sensor Over-range / Submerged Fault'; triggers sensor error alert.	Over-range fault alert triggered.	**Pass**
TC-007	TS-03	Boundary: Water level reaches exactly 80.0% bank capacity.	Bank Capacity: 10m Current Level: 8.0 m	Advisory warning triggered: '80% river capacity reached. Advisory active.'	Advisory SMS & Push alert dispatched.	**Pass**
TC-008	TS-03	Boundary: Water level reaches 100.0% bank full capacity.	Bank Capacity: 10m Current Level: 10.0 m	High-priority alert: '100% capacity reached. Evacuation mode enabled.'	Sirens activated & Evacuation SMS dispatched.	**Pass**
TC-009	TS-04	Normal/Rule 1: Rapid water rise (>0.6 m/min) with heavy rain.	Rise Rate: 0.7 m/min Rain: 65 mm/hr	System triggers 'Flash Flood Threat Alert' to local disaster teams.	Flash Flood alert generated.	**Pass**
TC-010	TS-04	Exceptional/Rule 2: Sudden surge due to dam release without local rain.	Rise Rate: 1.2 m/min Dam Gate: OPEN	Triggers Critical Dam Surge Alert and auto-opens downstream flood barriers.	Emergency barrier deployment initiated.	**Pass**
TC-011	TS-05	Normal: Admin triggers remote barrier gate opening command.	State: CLOSED Action: 'OPEN_GATE' Auth: Valid PIN	Gate state transitions from CLOSED → OPENING → OPEN in <3.0s.	Gate confirmed OPEN in 2.2s.	**Pass**
TC-012	TS-05	Exceptional: Gate fails to open due to mechanical jam.	State: CLOSED Action: 'OPEN_GATE' Ack: Timeout	Gate state transitions to ERROR; alert sent to Mechanical Maintenance Team.	State changed to ERROR and ticket logged.	**Pass**
TC-013	TS-06	Boundary (Zone 1): Evacuation dispatch for high-risk radius (2 km).	Distance: 2.0 km Threat: Critical	High Priority Evacuation SMS sent to all connected devices in Zone 1.	Zone 1 SMS broadcast completed.	**Pass**
TC-014	TS-06	Boundary (Zone 2): Advisory dispatch at zone boundary (15 km).	Distance: 15.0 km Threat: Warning	Advisory alert sent to Zone 2 residents; shelter centers alerted.	Zone 2 advisory dispatched.	**Pass**
TC-015	TS-06	Normal (Zone 3): Regional informational update (>15 km).	Distance: 25.0 km Threat: Low	Informational bulletin issued to Zone 3 civic authorities.	Zone 3 bulletin sent.	**Pass**
TC-016	TS-07	Invalid/Security: SQL injection payload in sensor query parameter.	GET /api/v1/sensor?id=' OR '1'='1	Server rejects request with HTTP 400/403; no internal DB schemas exposed.	HTTP 400 Bad Request; SQLi blocked.	**Pass**

5. REQUIREMENTS TRACEABILITY MATRIX (RTM) & TEST COVERAGE

Req ID	Requirement Description	Scenario ID	Test Case IDs	Coverage Level
FR-01	User Authentication & RBAC	TS-01	TC-001, TC-002	100% (Positive, Negative, Boundary)
FR-02	IoT Telemetry Ingestion	TS-02	TC-003, TC-004, TC-005, TC-006	100% (Normal, Low BVA, Neg BVA, Surge)
FR-03	Threshold & Flood Alerting	TS-03	TC-007, TC-008	100% (80% Advisory, 100% Evacuation)

Req ID	Requirement Description	Scenario ID	Test Case IDs	Coverage Level
FR-04	Flash Flood Detection	TS-04	TC-009, TC-010	100% (Flash flood, Dam release surge)
FR-05	Automated Gate Control	TS-05	TC-011, TC-012	100% (Normal Open, Mechanical Fault)
FR-06	Zone Threat Mapping	TS-06	TC-013, TC-014, TC-015	100% (Zone 1 High, Zone 2 Med, Zone 3 Low)
NFR-02	Security & SQL Injection Prevention	TS-07	TC-016	100% (Input sanitization & API protection)